

ALI KASSAB

Unity Game Developer

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SUMMARY

Gameplay and Systems Engineer with experience shipping cross-platform titles on PS5, PS4, PC, Android, and iOS. Specialized in scalable gameplay architecture, multiplayer systems, AI behavior design, and performance optimization for production-ready indie, AA, and AAA projects. Strong focus on clean architecture, modular systems, and long-term maintainability.

EDUCATION

Dual Degree Program

Oct 2020 - May 2024

Ain Shams University

Bachelor of Computer and Information Sciences

University of East London

Bachelor of Science, Software Engineering Program

SKILLS

Core Technologies:

Unity, C#, C++, SFML
Photon PUN, Photon Fusion
Console, PC & Mobile
Deployment

Systems & Architecture:

OOP, Design Patterns
System Architecture
Multithreading
Async Programming

Gameplay & Optimization:

Behavior Trees, FSM
Pathfinding (A*, BFS, DFS),
Multiplayer & Networking
Performance Optimization

PROFESSIONAL EXPERIENCE

Game Programmer [Mega Cat Studios, Inc.](#)

November 2024 – Present

Pittsburgh, Pennsylvania, USA

- Architected core systems for the [Pit of Agonies](#) mode in [God of War Sons of Sparta](#) including reward seeding, items, shop UI, and progression, reworking legacy systems to support the mode.
- Developed modular NPC/Enemy behavior systems using Behavior Trees and FSMs, supporting 10+ archetypes across combat, patrol, and interaction for [God of War Sons of Sparta](#).
- Implemented core gameplay feel systems (physics, camera, input, world-space UI) and a state-driven animation sequencing system for gameplay and cinematics in [Backyard Baseball](#).
- Contributed to C++ porting pipelines for Android, supporting 5 shipped [Backyard Sports](#) titles with performance and feature parity.
- Refactored and maintained internal engine libraries (multithreading, async, physics, networking), improving stability and long-term maintainability.
- Built a data-driven gameplay balancing system using YAML configs, enabling designer-driven tuning without code changes for [God of War Sons of Sparta](#).
- Decoupled input and gameplay systems in [Maximum Apocalypse](#), improving modularity and reducing cross-system dependencies

Unity Game Developer [Genesis Creations S.A.E](#)

August 2023 – July 2024

Cairo, Egypt

- Designed core game systems (error handling, notifier systems, UI, user access, command prompting) for the published mobile title [Sinai Heroes](#).
- Built VR hand pose tracking and object interaction features, increasing player immersion by 40%.
- Integrated networking, multiplayer, and matchmaking with Photon PUN, reducing connection issues by 70%.
- Optimized environments and meshes, reducing draw calls by 75%.

PROFESSIONAL EXPERIENCE

[Table For None](#) (Unity, C#)

April 2026

Multiplayer Detective Co-op | Lead Programmer | Team of 3

- Implemented multiplayer functionality using Unity Netcode.
- Designed major game systems and led the team to ship a successful early development trailer.
- Developed customer AI using FSM and a flexible cooking actions system.

[FakeOut BOSS](#) (Unity, C#)

February 2025

[GDA JAM](#) | 1st Place

Stealth, Simulation, and Comedy | Lead Programmer | Team of 3

- Engineered an NPC awareness system, where AI-controlled coworkers detect suspicious behavior based on activity, movement, and reaction time, increasing challenge adaptation by 35%.
- Integrated an adaptive time dilation mechanic, allowing players to slow time for precise decision-making during high-risk moments, improving last-second reaction success by 50%.
- Constructed an unpredictable NPC patrol, customizable pathfinding system (BFS, DFS, A* and its variants) and gaze system, preventing static behavior by introducing randomized state transitions, fake-outs, and varying alert levels, increasing replayability.
- Optimized task-balancing mechanics, forcing players to manage both in-game progress and workplace deception, leading to an increase in player engagement and multitasking efficiency.

[Mystic Sands](#) (Unity, C#)

September 2024

[ITI-BelMasry Tech Hackathon](#) | 1st Place

Narrative-Driven Adventure | Lead Programmer | Team of 4

- Expanded maze regeneration using Recursive Backtracking, improving efficiency by 40%.
- Integrated graph-based dialogue and quest systems, boosting engagement by 50%.
- Created realistic NPC AI with NavMesh pathfinding, increasing responsiveness by 50%.
- Structured modular UI with Command Design Pattern, reducing update time by 20%.

[Dungeon Of The Skeletons](#) (SFML, C++)

May 2021

2D Platformer | Developer | Team of 6

- Implemented UI and input systems.
- Built player character animation and movement systems.
- Architected collision systems to handle jump states and prevent overlapping.
- Designed collectibles and progression via quest items.
- Programmed hazards and enemies with patrol and turret behaviors.
- Managed scene/state transitions for menus, controls, and cutscenes.